

Zenish Borad

Blockchain & Backend Engineer

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Summary

Blockchain and backend engineer focused on systems whose results can be **independently verified**: Bitcoin full-node RPC pipelines into normalized SQL, Solidity contracts with full-coverage test suites, and an on-chain commitment layer that lets any third party replay a decision and confirm it was never altered. Works across the stack — Python data pipelines, smart contracts, and the React/Ethers interfaces that expose them.

Technical Skills

Blockchain & Web3: Bitcoin Core RPC, Ethereum EVM Internals, Solidity Smart Contracts, Uniswap v2/v3 AMMs, ERC20, Price Oracles, keccak256 Commitments & ECDSA Signing, On-Chain Data Modeling, Cryptography

Programming Languages: Python, SQL, JavaScript, Solidity, Rust, R

Backend & Web3 Tools: REST APIs, Flask, Web3.py, Ethers.js, Hardhat, React, CCXT, LLM Text-to-SQL, TA-Lib

Data & Quantitative: Pandas, NumPy, Scikit-learn, Feature Engineering, Predictive Modeling, Backtesting

Cloud & Infrastructure: AWS, Docker, Git, SQLite, PostgreSQL

Education

Northeastern University

Master of Science in Information Systems — GPA: 3.61 / 4.0

Sep 2025 – Expected May 2027

Boston, MA

Parul University

Bachelor of Technology in Computer Science and Engineering

2020 – 2024

Vadodara, India

Projects

ChainProof: Verifiable Trading Agent

Python, Flask, SQLite, Solidity, Hardhat, Web3.py

Jan 2026 – Aug 2026

- Shipped an on-chain verification layer on **Ethereum Sepolia**: hashed and signed 30 decisions with **keccak256** and **ECDSA**, committed them through a **verified Solidity contract**, and achieved **30/30** independent re-verification by replaying the unmodified strategy against historical data
- Built automated **tamper detection** that re-executes the strategy from scratch and cross-checks recomputed hashes against on-chain commitments, correctly flagging falsified records; two live dashboards expose verification status in real time
- Implemented the underlying decision engine in Python — **7 technical indicators** (RSI, MACD, Bollinger Bands, SMA/EMA, Stochastic, OBV) under a **4/7 majority vote**, with automated risk controls (3% stop-loss, 20% take-profit, 10% daily loss limit); **+38.47%** backtested on BTC-USD

Bitcoin Text-to-SQL Analytics Pipeline

Python, Bitcoin Core RPC, SQLite, Gemini API

- Built a pipeline syncing **48,000+ transactions** from a **Bitcoin full node** into a normalized SQLite schema spanning blocks, transactions, inputs, outputs, and addresses
- Layered a **natural-language-to-SQL engine** (Gemini API) over that schema, returning correct SQL on **75% (9/12)** of a self-designed query benchmark

SimpleAMM: Constant-Product AMM & Web3 UI

Solidity, Hardhat, React, Ethers.js

- Implemented a **constant-product AMM** in Solidity with a 0.30% swap fee and transferable **ERC20 LP shares**, backed by a Hardhat test suite at **100% line and branch coverage**
- Deployed a live **React/Ethers.js** interface on Sepolia — deposit, redeem, and swap flows, a live $x \cdot y = k$ reserves curve, and swap price history reconstructed from **on-chain event logs via eth_getLogs**

Experience

Sigzen Technologies Pvt. Ltd.

Backend Engineer Intern

2024

Ahmedabad, India

- Consolidated inventory and sales data from **three separate business systems** into a single normalized dataset using **Python (Pandas, NumPy)**, replacing a manual spreadsheet process repeated every reporting cycle
- Containerized the workflow with **Docker** so the same pipeline ran identically across development and production, making scheduled runs reproducible rather than machine-dependent
- Built **SQL dashboards** over the unified dataset, giving product and finance teams direct query access to trend data they had previously assembled by hand